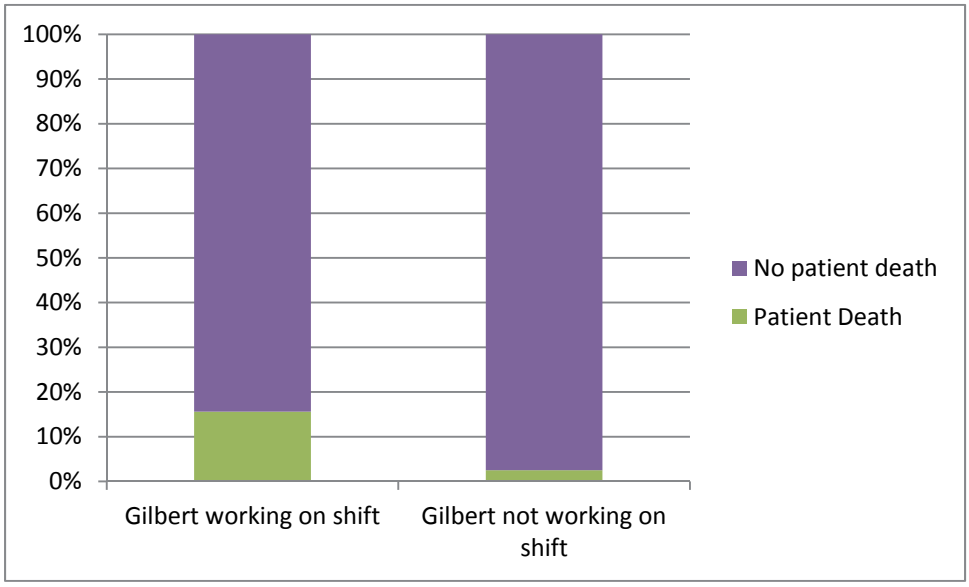
Solutions to Exploration 5.1

1. Obs units = 1,641 shifts
2. Whether or not Gilbert worked on the shift = categorical; whether or not a patient died during the shift = categorical.
3. Explanatory = whether or not Gilbert worked; response = whether or not a patient died.
4. Observational study. The researchers are not assigning the values of either variable.
5. We need to cross-tabulate the data. That is, we need to know how often each possible co-occurrence of outcomes actually happened; namely, how often Gilbert worked and someone died, how often Gilbert worked and no one died, how often Gilbert didn't work and someone died, and how often Gilbert worked and no one died.
6. The counts are not helpful without knowing the conditional proportions.
7.

	Gilbert worked shift	Gilbert did not work shift	Total
Patient died during shift	40	34	74
No patient died during shift	217	1,350	1,567
Total	257	1,384	1,641

8. Compare the "death rate" when Gilbert was working, 40/257 (15.6%), to when she wasn't working, 34/1,384 (2.5%). This is more informative than comparing the counts, which don't account for the fact that Gilbert worked less than 20% of the time.
9. a) 15.6% (40/257)
   b) 2.5% (34/1,384)

10.



11. Yes, the death rate is quite a bit higher when Gilbert works than when she doesn't work.
12. It seems large enough to be convincing.
13. The difference in proportions
14. (15.6% – 2.5% = 13.1%); that is a fairly large difference.
15. 15.6/2.5 = 6.24. The patient death rate is more than six times higher when Gilbert is working as compared to when she is not working. This does seem like quite a large ratio.
16. No! For example, if Gilbert worked the night shift and more people die during the night shift (or holidays or weekends or whatever) that could explain it.
17. See answer to #16.
18. a) 38.9% death rate when Gilbert worked; 25.8% death when Gilbert didn't work.

    b) 38.9% – 25.8% =13.1%—same as we had before

    c) Ratio = 38.9/25.8 = 1.51 times as much. This ratio is much different than we had before.

    d) Even though the differences are the same, the evidence is more convincing in the first table, which had the overall lower death rate.

19. In this table, the conditional proportions are approximately 10% in both cases (25/257 and 138/1,384).

	Gilbert worked shift	Gilbert did not work shift	Total
Patient died during shift	72	385	457
No patient died during shift	185	999	1,184
Total	257	1,384	1,641

20. The bar heights would be approximately equal.

Solutions to Exploration 5.2

1. a) A control group is necessary because this helps us account for the fact that some people will yawn even without the "yawn seed."

   b) To control for confounding variables; to be able to conclude that the yawn seed is what "caused" the difference in yawning percent

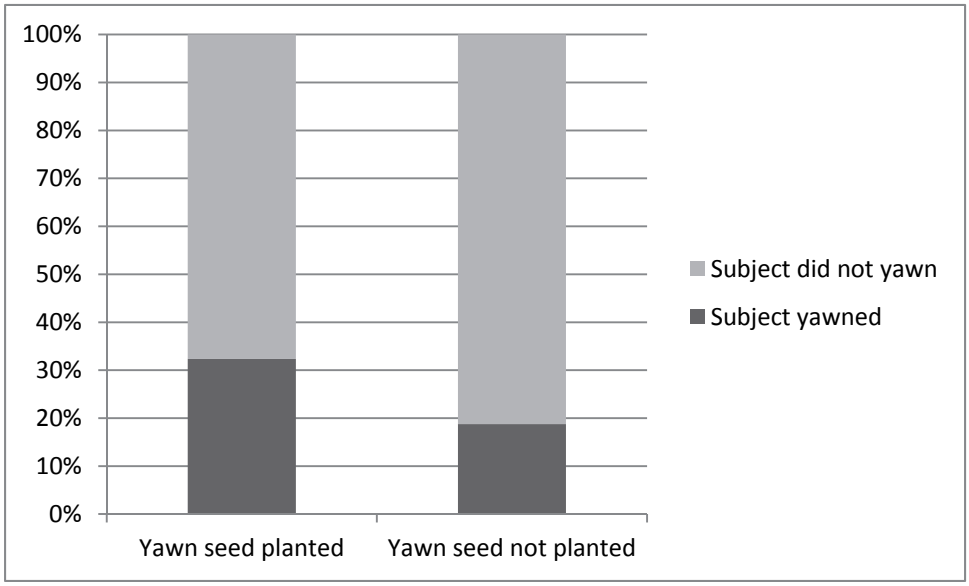
   c) Randomized experiment. The researchers randomly assigned the treatment (yawn seed or no yawn seed).

   d) No, probably just got volunteers. To do a random sample they would have to, for example, make a list of all the people at the flea market and randomly choose some.
2. Explanatory (yawn seed = yes/no); response (patron yawned = yes/no). Both variables are categorical.
3. Null: Yawning is not contagious. Alt.: Yawning is contagious.
4. No association
5. There is an association.
6. Null: There is no association between whether or not someone gets the yawn seed and whether or not the person yawns. Alt: There is an association between whether or not someone gets the yawn seed and whether or not the person yawns.
7. Null: The long-run probability someone yawns when they see the yawn seed. Alt: The long-run probability someone yawns when they don't see the yawn seed.
8. Null: The long-run probability someone yawns when they see the yawn seed is the same as when they don't. Alt: The long-run probability someone yawns when they see the yawn seed is different than when they don't.
9.

	Yawn seed planted	Yawn seed not planted	Total
Subject yawned	11	3	14
Subject did not yawn	23	13	36
Total	34	16	50

10. Proportion who yawned in treatment group: 11/34 = 32.4%; proportion who yawned in control group: 3/16 = 18.8%. Difference is 32.4% – 18.8% = 13.6%.

11.



12. a) Yes.

    b) Not much, the difference in proportions is not very large.

13. The difference occurred by chance (yawn seed doesn't matter) or the difference did not occur by chance (yawn seed does matter).

14. (*Note:* Answers will vary.)

    a)

	Yawn seed planted	Yawn seed not planted	Total
Subject yawned	11	3	14
Subject did not yawn	23	13	36
Total	34	16	50

    b) 11/34 = 32.4%, 3/16 = 18.8%, differences is 32.4% – 18.8% = 13.6%

    c) It is the same as our observed statistic.

15. a) 9 in seeded group. Difference is –0.048.

    b) 0.044, 0.044, –0.048, –0.140. Not the same each time. Each simulation has the potential to yield a different result.

16. Null hypothesis: There is no difference in the probability of yawning between the two groups. One repetition = one shuffle and deal of the cards. Statistics = difference in proportions (or number of yawners in each group).
17. a) Yes. The null hypothesis suggests that the difference in proportions will be zero.

    b) No, 0.136 is not out in the tail of the distribution. 0.136 is a typical result.

    c) 0.136 or more

18. 0.254 is the approximate p-value.
19. Under the assumption that yawning is not contagious, if we repeated the shuffling/dealing many, many times, the probability we would obtain results as or more extreme than 0.136 is 0.254.
20. No, the *Mythbusters* result does not provide much evidence that yawning is contagious.
21. 10/34 = 29.4%, 4/16 = 25%, differences is 29.4% – 25% = 4.4%
22. Smaller
23. Larger, weaker. A smaller difference in proportions is closer to zero, the center of the null distribution, so the p-value (proportion of simulated statistics as extreme) will be larger; larger p-values offer weaker evidence against the null.
24. Approximate p-value = 0.5192; this p-value provides very weak evidence against the null and we can conclude it is plausible that yawning is not contagious.
25. a) The difference in the probability of yawning based on whether or not someone observed a yawn "seed"

    b) 0.044

    c) 0.138

    d) 0.044 ± 0.276, (–0.232, 0.320)

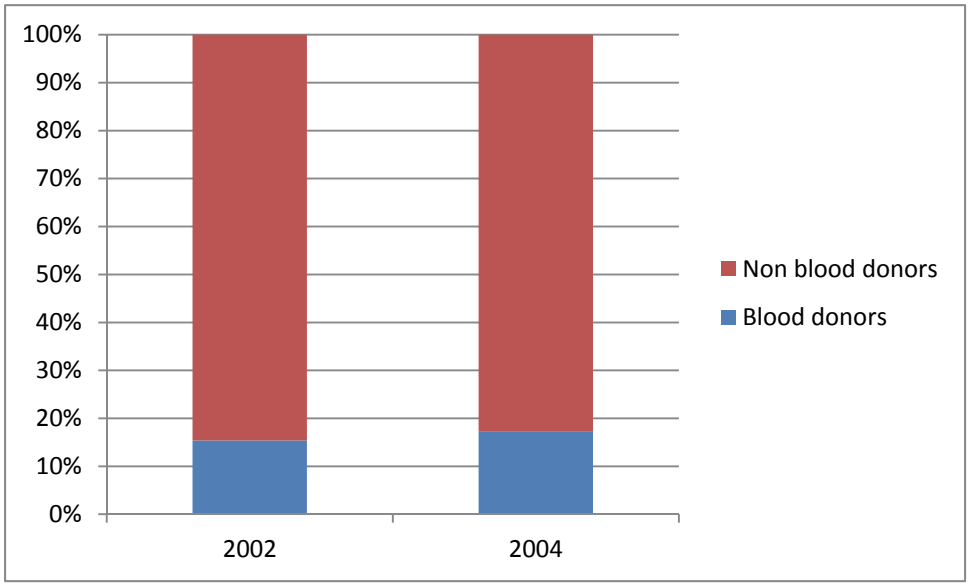
    e) Both positive and negative values

    f) Since 0 is in the interval, 0 is a plausible value for the difference in proportions, which does not provide evidence of a difference in the proportions.
26. No, while it is true that a higher proportion (29%) of people yawned when viewing the yawn "seed" than the proportion of yawners without a yawn seed (25%), the difference (4.4%) is very small and happens quite easily just by random chance. The simulation we conducted helped to quantify that a difference like 4.4% (or larger) happens about 50% of the time just by chance (even when yawning is NOT contagious).
27. Yes, because it is a randomized experiment.

28. It's not a random sample, so, generalizing the result very much is not a great idea. We're not even sure if the people in the study are a good representation of people at the flea market since the participants in the study may be systematically different than the people not in the study.
29. The two conclusions are not the same because (i) suggests that we've proven yawning is not contagious while (ii) suggests that our study didn't prove yawning was contagious. (ii) is the more appropriate conclusion
30. Type II error, missed opportunity. We failed to reject the null hypothesis, and the alternative hypothesis could have been true. We may have missed an opportunity to confirm the commonly held notion that seeing people yawn increases the chances that we will yawn ourselves.
31. Connecting a random sample of people (for example, randomly choosing people and going door to door) would improve our ability to generalize to a larger population. A bigger sample size is likely needed. Plus, we need to be sure that the yawn seed is both realistic (not obviously fake) and noticed by the respondents and that adequate time is given to see if the person yawns. Asking participants if they "felt like they had to yawn" (if they didn't yawn) might also be interesting, since many people might be stifling the yawn to be polite.
32. a) Same
    b) Smaller
    c) Stronger
    d) Fail to reject null
33. The p-value = 0.187; weak evidence against the null at the 0.05 significance level
34. We now have stronger evidence that yawning is contagious but the evidence is still not very strong.
35. 1
36. Null: The relative risk of yawning with a yawn seed to yawning without a yawn seed is 1.
    Alt: The relative risk of yawning with a yawn seed to yawning without a yawn seed is greater than 1.
37. 1.725
38. They are the same.

Solutions to Exploration 5.3

1. Random sampling only; means we can generalize to the population of all adult Americans
2. Null: The proportion of all U.S. adults who gave blood in 2002 is the same as in 2004.
   Alt: The proportion of all U.. adults who gave blood in 2002 is different than in 2004.
   Null: $\pi_{2002} - \pi_{2004} = 0$, Alt: $\pi_{2002} - \pi_{2004} \neq 0$
3. 2002 proportion: $\hat{p} = 15.4\%$, 2004 proportion: $\hat{p} = 17.2\%$, difference (2002–2004) : –1.8%
4. The proportion of donors in 2002 is very close to the proportion of donors in 2004.



5. Instructions for entering a table in the **Two Proportions** applet
6. a) The proportion of simulated differences in proportions smaller than –0.018 or larger than 0.018
   b) p-value = 0.19
   c) We do not have evidence that the proportion of blood donors in 2002 is different than in 2004.
7. Yes, approximately normal; mean = 0, SD = 0.014
8. a) –0.018
   b) –0.018/0.014 = –1.29
   c) Yes, the standardized statistic is consistent—the standardized statistic is not very large (less than –1.5 or larger than 1.5), so we do not have evidence of a difference.
9. The normal distribution overlays fairly well with the simulated null distribution. The p-value = 0.2058, similar to what we got before.
10. a) (–0.046, 0.01)

b) We are 95% confident that the parameter (difference in true proportion of blood donors in 2002 and 2004) is between –0.046 and 0.01.

c) The interval contains both positive and negative values—this means that 0 is a plausible value for the difference in the two proportions—in other words, we don't have evidence to reject the null hypothesis that the two proportions are the same.

11. Calculations verified; same p-value and confidence interval
12. Yes, 210, 1152, 230, and 1106 are all larger than 10—thus the validity conditions are met.
13. Yes, because the data are from a random sample of Americans.
14. a) Standardized statistic would switch signs, p-value would remain the same, and confidence interval would be same width, but centered at 0.018 instead of -0.018.

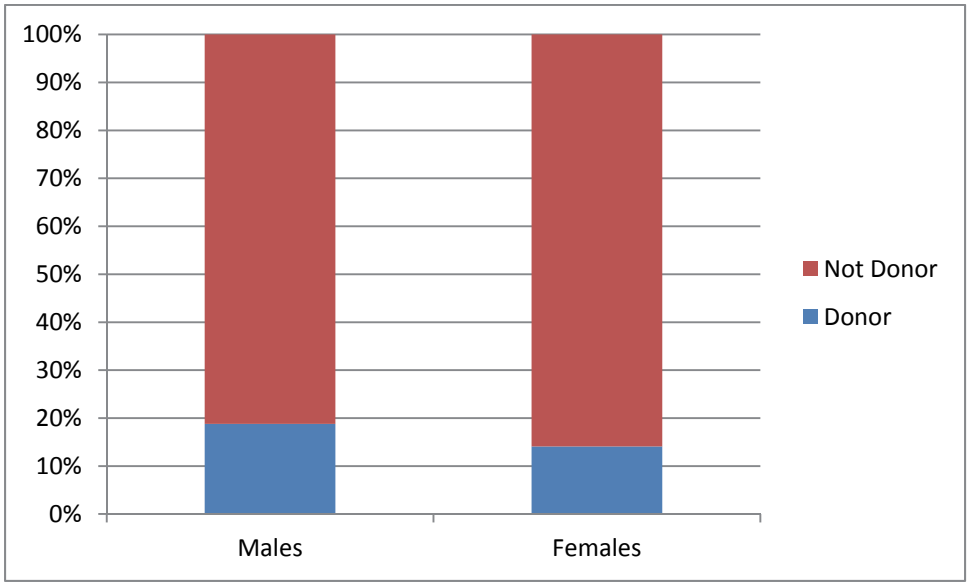
    b) 1.27, p-value = 0.21, confidence interval (–0.01, 0.046)

    c) Yes!
15. The validity conditions are met (239, 1,032, 201, and 1,226 are all larger than 10), so we'll use the theory-based approach.

    Null hypothesis: The proportion of U.S. male blood donors is the same as the proportion of U.S. female blood donors.

    Alt hypothesis: The proportion of U.S. male blood donors is different than the proportion of U.S. female blood donors.

    The proportion of male blood donors is 239/1,271 = 18.8%, for females is 201/1,427 = 14.1%



Standardized statistic = 3.31, p-value = 0.0009.

We have very strong evidence that the proportion of male blood donors is different than the proportion of female blood donors. In particular, we are 95% sure that males donate blood between 1.9% and 7.5% more often than females in the United States.